LEAKLISTEN · DESIGN DECISION RECORD

LeakListen recommendations accepted

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DOCUMENT CONTROL

Document ID	LKL-DDR-002	Version	0.2
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/leaklisten		
Source	docs/decisions/0002-recommendations-accepted.md @ a81e84a		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	O1 and O2 decided by Amish on 2026-10-02 (LKL-DEC-001, items 4 and 5)
0.1	2026-09-25	Amish Chadha	Record the recommendations accepted by Amish on 2026-09-25, what changed in the repo and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Every item below marked "Decided by Amish, 2026-09-25: go with recommendation" is a decision by Amish. Items without a recommendation stayed "Proposed, awaiting Amish" at this record; O1 and O2 were decided by Amish on 2026-10-02 (LKL-DEC-001, items 4 and 5): "i approve your recommendations for all 555 open decisions."

Context

On 2026-09-25 Amish wrote, in chat: "i accept all your recommendations, go with them across all repos." Before that, LeakListen carried two groups of items: D1 to D6 in LKL-DDR-001, adopted as recommended for TRL 3 and open for his review, and three new items raised by the TRL 3 calculations in `docs/REVIEW.md` (radio under iron covers, plastic mains, a larger magnet), each with a recommendation. Two items (O1, O2) never had a recommendation. TRL 4 remains on hold by Amish's instruction, so any part of a decision that needs a build, test, measurement or trial is recorded as decided but on hold.

Options considered

The options for D1 to D6 are in LKL-DDR-001 and the TRL 2 section of `docs/REVIEW.md`. The options for D7 to D9 are in the TRL 3 section of `docs/REVIEW.md` ("Still awaiting Amish", items 3 to 5) and in LKL-CAL-001 v0.1, sections D, G and I.

Decision

Table 1. Items decided by Amish on 2026-09-25.

#	Item	Decision (the recommendation)	Status	What changed in the repo
D1	Screening versus correlation	Nightly noise-level screening; correlation is a later variant	Decided by Amish, 2026-09-25: go with recommendation	Already reflected at TRL 3; LKL-DDR-001 v0.2 status wording
D2	Radio from under covers	Flat antenna under the cover, with the link budget deciding whether more is needed	Decided by Amish, 2026-09-25: go with recommendation	Already reflected; the link budget's answer is D7
D3	Sensor type	Piezo disc with a charge amplifier; hydrophone kept for plastic networks	Decided by Amish, 2026-09-25: go with recommendation	Already reflected; D8 gives the hydrophone variant its requirement
D4	FieldNode radio core	Reuse the STM32WL core and payload conventions, without FieldNode's solar power	Decided by Amish, 2026-09-25: go with recommendation	Already reflected; no FieldNode change needed
D5	Primary cell size	C size, about 7.7 Ah	Decided by Amish, 2026-09-25: go with recommendation	Already reflected
D6	Privacy rule	Raw samples never leave the device; R8 fixed	Decided by Amish, 2026-09-25: go with recommendation	Already reflected
D7	Radio under iron covers (R7)	Measure cover loss first; then plan a gateway within about 0.5 km of each district with iron covers as the default, with a through-cover antenna or composite cover where a utility agrees	Decided by Amish, 2026-09-25: go with recommendation. The cover loss measurement is TRL 4 work and on hold	R7 restated in LKL-REQ-001 v0.4 against a gateway within 0.5 km under iron covers (1 km where a composite cover or through-cover antenna is agreed); LKL-CAL-001 v0.2 adds the 0.5 km margin [D5]: +0.9 dB EU868 SF12, -1.2 dB US915 SF9. R7 moves from not met to at risk. Gateway density raised with TwinKit as a cross-repo action
D8	Plastic mains (R2)	Restate the contact sensor's scope as metallic mains; the hydrophone variant (D3) is the answer for plastic networks	Decided by Amish, 2026-09-25: go with recommendation. Sizing the hydrophone variant is later TRL 3 work, not done here	R1 and R2 restated in LKL-REQ-001 v0.4; R2 moves from not met to open (variant not sized). Pitch and problem lines unchanged, since hydrophones also mount on existing fittings
D9	Magnet (R9)	Fit a 42 mm pot magnet, about 600 N rated, with a keeper plate	Decided by Amish, 2026-09-25: go with recommendation	`cad/src/model.py` magnet 32 to 42 mm; STEP and STL re-exported; `bom/bom.csv` line 2 $5 to $9, total $109.00 to $113.00; LKL-DWG-001 Rev P1 to P2; LKL-CAL-001 v0.2: hold at a 0.5 mm coating 60 N to 125 N (R9 at risk to met on paper), mounted resonance 776 to 1,553 Hz to 695 to 1,389 Hz, mass 0.93 to 0.98 kg; LKL-PRC-001 v0.4 and media re-rendered

Table 2. Items still open.

#	Item	Status
O1	First partner utility and region for co-design and a pilot district. No recommendation was made.	Decided by Amish, 2026-10-02 (LKL-DEC-001, item 4): a water utility with metallic mains, district metered areas and a leakage target; first candidate region to approach, the UK (radio band EU868). Nothing is agreed.
O2	Whether a vibration calibration check belongs in LeakListen or in CalRig. No recommendation was made.	Decided by Amish, 2026-10-02 (LKL-DEC-001, item 5): a small shaker bay in CalRig, with LeakListen defining the acceptance test (frequency range, level and pass band) in its own repo.

Consequences

- `project.yaml` : no budget, pitch or problem change was recommended. `budget_usd` stays at $130; parts cost is now $113.00, a margin of $17.00. `trl` and `trl_target` stay at 3.
- Requirement status (LKL-CAL-001 v0.2): none not met (was 2), 4 at risk (R1, R3, R4, R7), 1 open (R2), 2 not verifiable at TRL 3 (R10, R13), 5 met on paper (R5, R6, R9, R12, R15) and 3 met by design (R8, R11, R14).
- R3 gets slightly worse with the heavier magnet (flat to about 376 to 752 Hz instead of 420 to 840 Hz), and R12 keeps only 20 g of margin.
- Documents revised: LKL-PRB-001 v0.4, LKL-PRC-001 v0.4, LKL-REQ-001 v0.4, LKL-CAL-001 v0.2, LKL-DDR-001 v0.2, LKL-DWG-001 Rev P2.
- Cross-repo action: TwinKit's gateway sizing should allow for about 0.5 km spacing in districts with cast-iron covers. Recorded in `docs/REVIEW.md` ; TwinKit is not edited here.
- On hold (TRL 4): measuring cover loss through cast-iron and composite covers, and any field check of the link. Nothing in this record authorizes building, testing or purchasing.